A Novel Transformer-KAN Network Utilizes Multimodality Contrastive Learning and Mask Reconstruction for Remote Sensing Classification

Shuxiang Xia, Xiaohua Zhang, *Member, IEEE*, Hongyun Meng, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Contrastive learning is a promising paradigm in the multimodal few-shot domain, leveraging images captured from different sensors as diverse perspectives of the same ground objects. However, existing methods encounter two major challenges: relatively weak discriminative ability, complex features inherent in remote sensing data, and insufficient cross-modal interactions, as these methods often align features within individual modalities without fully exploiting their complementary information. These limitations hinder the overall performance of models in multimodal remote sensing tasks. In this article, we propose a novel framework to address these challenges. First, we introduce the Transformer as the encoder, and at the same time, we replace traditional multilayer perceptrons (MLPs) with the Kolmogorov–Arnold network (KAN) for feature dimensionality reduction. The model reduces parameters slightly without affecting accuracy, achieving better efficiency. Second, we design a specialized projection layer to align and deeply integrate spatial and light detection and ranging (LiDAR) modalities, effectively harnessing their complementary strengths and implementing a self-supervised decoding mechanism to reconstruct masked image patches generated during the data augmentation process, enforcing spatial and structural learning for robust multimodal representations. Extensive experiments conducted on four public datasets demonstrate the superior classification performance of our proposed method, outperforming state-of-the-art approaches and confirming its efficacy in multimodal remote sensing tasks.**

***Index Terms*— Autoencoder, contrastive learning, hyperspectral image classification, light detection and ranging (LiDAR), self-supervised learning, Transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) provides nearly continuous spectral information within a specific wavelength range, enabling the analysis of material chemical compositions and the identification of different land cover types through variations in spectral reflection curves [1], [2], [3]. However, HSI data lack the capability to capture 3-D geometric features such as target height and volume and are highly susceptible to weather conditions (e.g., rain and haze), which can degrade the accuracy of spectral reflections. Consequently, the classification accuracy significantly decreases when distinguishing between land cover types with identical chemical compositions but different structures or when data are collected under adverse weather conditions. In contrast, light detection and ranging (LiDAR) can obtain high-precision 3-D structural information of land features by measuring the reflection intensity and time delay of laser pulses [4], [5], [6]. LiDAR is less affected by external environmental stability and can differentiate targets with similar chemical compositions but different geometric structures. However, LiDAR lacks of in-depth perception of an object's material information. Therefore, HSI and LiDAR data exhibit strong complementary characteristics, and their fusion can enhance classification accuracy in applications such as agricultural and forestry monitoring [7], urban planning and building detection [8], as well as geological exploration and environmental assessment [9]. This complementary advantage has been increasingly validated in numerous practical applications [10], [11].

Early studies predominantly relied on manual feature engineering techniques to fuse hyperspectral (HSI) and LiDAR data. For instance, features were individually extracted from each modality using methods such as extended morphological profiles (EMPs) [12] and independent component analysis (ICA) [13], followed by fusion operations like concatenation, addition, or weighting. However, these approaches generally lack a thorough exploration of cross-modal correlations and often require manual configuration of fusion parameters or weight distributions. As a result, they struggle to adaptively balance the contributions of different modalities across diverse land cover types, potentially leading to information redundancy or loss. To address these limitations, deep learning-based approaches have gained substantial traction in recent years. Through end-to-end gradient-based training, deep learning models can automatically learn robust multimodal feature representations with superior generalization capabilities [14], [15]. For example, Huang et al. [16] utilized multiscale 3-D–2-D convolutions to independently extract

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HSI and LiDAR features, aiming to alleviate the "over-smoothing" problem encountered in traditional graph neural networks (GNNs) during high-order aggregation. Yao et al. [17] proposed a deep hybrid multigraph (DHMG) neural network that leverages spectral and auto-regression and moving average model (ARMA) filters to extract spectral features while suppressing graph noise. Wang et al. [18] introduced a learnable Transformer architecture with an adaptive gating mechanism and learnable transformer (AGMLT) to mitigate the structural flattening issue caused by sample preprocessing. Yu et al. [19] proposed a hyperspectral intrinsic image decomposition-based fusion network that integrates horizontal geometry from shading and vertical geometry from LiDAR to enhance the geometric correlation and interpretability of multimodal remote sensing data. Xue et al. [20] developed a deep hierarchical vision Transformer framework to effectively integrate heterogeneous multimodal features. However, these supervised learning-based models still face several critical limitations in remote sensing tasks. First, they require large volumes of accurately labeled data to achieve satisfactory performance. In practice, labeling hyperspectral and LiDAR data is costly and time-consuming due to their high spectral dimensionality, spatial resolution, and modality heterogeneity [21], [22]. Second, models trained on imbalanced datasets are often biased toward majority classes, which is particularly problematic in land cover scenarios with rare or fragmented categories [23]. Moreover, traditional supervised training fails to generalize well when deployed on unseen areas or when new classes appear, making it difficult to scale. These factors jointly motivate the need for label-efficient alternatives such as few-shot learning (FSL), self-supervised learning, and contrastive representation learning, which can utilize limited annotations and improve generalization across modalities and scenes. To mitigate these challenges, FSL has emerged as a promising research direction [21], [22], focusing primarily on two key aspects.

First, design more effective network architectures to extract more discriminative features and reduce the sample requirement. Convolutional neural networks (CNNs) have demonstrated exceptional performance in the field of image recognition [23], [24], [25]. Their local receptive mechanisms effectively extract spatial–spectral features from local regions of HSI, while weight-sharing strategies significantly reduce the number of parameters, thereby ensuring good training efficiency and generalization capabilities in scenarios with high feature redundancy in HSI data. Ge et al. [26] proposed a spatial–elevation–spectral convolutional feature extraction module to enhance discriminative ability under diverse land cover conditions. However, CNNs are designed for Euclidean regular grid data and exhibit certain limitations when applied to irregularly distributed natural land covers; they exploit correlations among pixels within a limited scope. In contrast, GNNs take the entire image as input, constructing each pixel as a graph node, connecting adjacent or spectrally similar pixels within the graph, and aggregating multimodal information through message-passing mechanisms. Cai et al. [27] developed a graph-attention-based multimodal fusion (GAMF) network feature extractor to explore the correlations between different modalities. Shen et al. [28] integrated a hierarchical prototype alignment mechanism into GNNs to explicitly model semantic consistency and intrinsic similarity of the same land cover class across different scenes, thereby significantly enhancing the generalization capability of HSI classification across scenes. However, as the number of nodes increases, the computational cost escalates dramatically. Transformers, leveraging multihead self-attention (MSA) mechanisms, theoretically capture both local and global feature dependencies simultaneously [29], [30], [31]. Their parallelizable architecture and inherent ability to model long-range dependencies make Transformers more flexible in integrating contextual semantics of distant pixels and geometric information provided by LiDAR in HSI and LiDAR fusion tasks. To prevent the disconnection between spectral and spatial features when extracting spatial information from HSI, Yang et al. [32] proposed a modality fusion vision Transformer module, which utilizes rich spectral information while preserving essential spatial structures during spatial feature extraction. However, Transformers rely on global attention mechanisms and lack inductive biases, making it more challenging for models to learn structured features without sufficient supervision or reasonable prior information.

Second, propose more flexible learning strategies to exploit the intrinsic structure and distribution of limited labeled and abundant unlabeled samples. Specifically, data augmentation techniques expand the sample size by applying transformations such as rotation [33], cropping [34], and spectral masking [35] to a limited number of labeled samples. Nalepa et al. [36] designed a data augmentation method based on principal component analysis (PCA) for use during deep network training. However, these methods primarily rely on labeled samples and may lead to synthetic data deviating from the true distribution if the augmentation strategies are not appropriately chosen. Semi-supervised learning [37], [38] typically employs a "teacher–student" model framework: a teacher model is trained using a small amount of labeled data and then generates pseudo-labels for a large amount of unlabeled data to guide the training of the student model. Li et al. [39] proposed a novel label calibration module (LCM) that generates high-confidence pseudo-labels by comparing multiview classification based on superpixel segmentation to assist in the iterative optimization of semi-supervised models. Su et al. [40] designed a kernel-based iterative filter to capture global correlations in the manifold, aiming to address the problem that existing methods fail to exploit correlations in noncontiguous regions. Pu et al. [41] designed a semi-supervised multimodal dual-path network (SMDN) that collaboratively trains two unsupervised features mapped into a latent subspace under the guidance of a limited set of labeled training samples. However, if the teacher model lacks robustness, erroneous labels may accumulate during iterative processes, thereby degrading the final classification accuracy of the student model. Self-supervised learning, on the other hand, constructs pretext tasks based on the inherent attributes of the data to generate pseudo-labels, enabling the network to learn general discriminative features from large-scale unlabeled data [42], [43], [44], [45]. Depending on the specific tasks and classification targets,

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using labeled samples for fine-tuning helps enhance the specificity of the learned features. This approach can achieve effective representations even in scenarios with imbalanced class distributions. Existing studies include Jia et al. [46], who designed a multilevel convolutional fusion network for collaborative feature extraction for classification tasks; Wang et al. [47], who proposed bilinear attention convolution to fully exploit second-order and high-order feature interactions between HSI and LiDAR; and Xiu et al. [48], who utilized a multisource attention network combined with discriminative graphs and information entities to uncover pixel-wise topological relationships and improve collective classification. In these methods, hyperspectral data are often treated as a single modality, without adequately distinguishing or separately modeling its spatial and spectral information at the network input stage. In reality, the spatial and spectral dimensions provided by HSI differ significantly in both physical meaning and information form: the spatial modality focuses on the structure and texture of land cover in a 2-D plane or 3-D environment, emphasizing macro semantic information; the spectral modality reveals material composition and microscopic differences through continuous sampling of reflectance or radiance across various bands. Without proper distinction, the "same spectrum, different object" phenomenon in hyperspectral data can cause interference from spatial information corresponding to certain bands during classification tasks. Additionally, for tasks requiring LiDAR information fusion, treating hyperspectral data as a single modality can obscure the complementary nature of the internal "spatial–spectral" dual modalities and hinder the precise coupling with the elevation information provided by LiDAR. Addressing these issues, some studies have proposed dividing HSIs into "spatial modality" and "spectral modality" and introducing triplet contrastive learning to extract features separately [49]. However, this approach only aligns one-to-one correspondences between modalities and fails to sufficiently explore deep cross-view interaction mechanisms. For instance, when fusing spatial modalities with LiDAR data, it is necessary to model the dynamic relationships between semantic and geometric information at a higher level to achieve complementary gains.

Overall, Transformer-based multimodal remote sensing classification in FSL faces the following challenges.

1) Transformer-based models exhibit a quadratic increase in computational cost with sequence length. When dealing with high-resolution HSIs, this leads to excessive computational resource consumption, thereby limiting the model's applicability and scalability.
2) The Transformer relies on raw data to calculate pixel correlations but overlooks prior knowledge such as local connectivity and weight sharing. In HSI processing, factors such as noise and varying observation conditions make it difficult to maintain consistent similarity between adjacent pixels. This exacerbates the problem of the Transformer lacking inductive bias, thereby weakening its ability to effectively capture and utilize local semantic information through global similarity.
3) Existing methods that fuse spatial information with LiDAR data lack a deep cross-view interaction mechanism. As a result, the complementary information from different data sources is not fully exploited, which, in turn, limits the overall effectiveness of feature fusion.

To overcome these challenges, this article introduces the following innovations. In the feature extraction stage, a Transformer is employed as an encoder. Simultaneously, to mitigate the risk of feature dimensionality explosion resulting from the fusion of three modalities (spectral, spatial, and LiDAR information), this article proposes replacing traditional multilayer perceptrons (MLPs) with a Kolmogorov–Arnold network (KAN) network, thereby reducing the number of parameters while maintaining sufficient modeling capacity. Additionally, recognizing that Transformers, despite their ability to capture long-range dependencies, lack inductive biases, this study incorporates a decoder during training to reconstruct image patches obscured during data augmentation. This imposes explicit spatial structure and semantic context learning constraints on the network, enabling robust and discriminative multimodal representations even in unsupervised or self-supervised scenarios. Finally, to further exploit the complementary nature of spatial and LiDAR modalities, a specialized projection layer is introduced during the fusion phase to project these two modalities into a unified representation space for interaction, thereby achieving a dynamic fusion of semantic and geometric information at a higher level. In summary, the contributions of this article are as follows.

1) *Incorporating the KAN Network to Replace Traditional MLPs:* This substitution preserves modeling flexibility while reducing parameters slightly, especially under limited data conditions.
2) *Introducing a Decoder to Address the Lack of Inductive Bias in Transformers:* By reconstructing patches obscured during data augmentation, the network is explicitly constrained to learn spatial structures and semantic contexts, ensuring robust multimodal representations in unsupervised or weakly supervised settings.
3) *Implementing a Dedicated Projection Layer for Aligning and Interacting LiDAR and Spatial Modalities:* This further strengthens the comprehensive description and discriminative capability of the land cover types to be classified.

The rest of this article is organized as follows. Section II shows the details of our model. The experimental description and the result analysis on four public datasets are provided in Section III. Finally, Section IV summarizes the conclusions of this article.

II. Methodology

In this section, we present a comprehensive explanation of our Transformer-KAN-based tri-modal contrastive learning autoencoder method. This method conducts contrastive learning among the spatial, spectral, and LiDAR modalities and simultaneously performs modality-specific reconstruction. As illustrated in Fig. 1, for each modality, we apply masking at a certain ratio before feeding the data into the encoder. The extracted features are then projected into a feature space via a perceptron for contrastive learning, while those same features are simultaneously input into the decoder for modality reconstruction.

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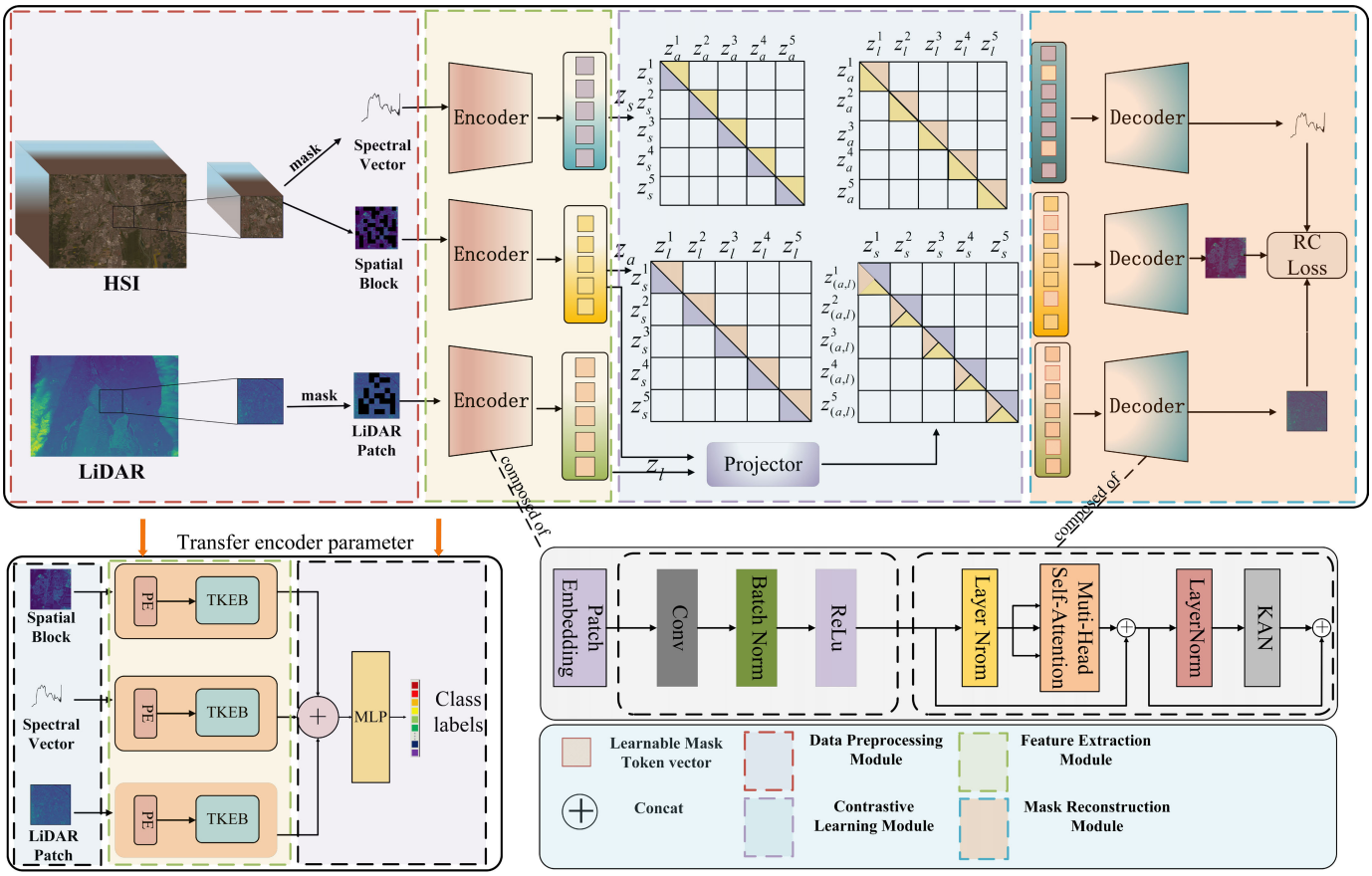


Fig. 1. Overall framework of the HSI and LiDAR classification model consists of two stages: pretraining and fine-tuning. In the pretraining phase, data augmentation (i.e., mask-based augmentation) is first applied to the different modalities. Subsequently, feature extraction is carried out through an encoder module composed of a patch embedding (PE) layer and a transformer encoder blocks based on KAN that incorporates KAN encoder blocks. To fully capture the complementary information between different modalities, a pairwise contrastive learning strategy is employed to model the relationships among the features extracted from each modality. Next, the extracted features are combined with a set of learnable vectors to reconstruct the original data sequence, and a decoder is used to reconstruct the masked portions. In the fine-tuning phase, the encoder parameters are transferred, and an MLP network is appended to perform land cover classification.

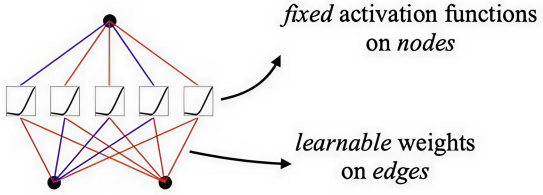


Fig. 2. Network architecture of KAN.

A. Overall Model Classification Process

We propose a self-supervised learning classification framework grounded in contrastive learning and reconstruction learning. In the pretraining phase, remote sensing data are preprocessed by splitting the HSIs into two modalities—spectral and spatial—and applying a masking strategy for data augmentation. Subsequently, we employ a Transformer-KAN encoder to extract features, where the KAN replaces the traditional MLP for feature dimensionality reduction, thereby reducing computational costs and mitigating overfitting. The architecture of KAN is illustrated in Fig. 2.

Next, contrastive learning is utilized to enhance cross-modal feature fusion by enforcing consistency in feature representations through multiple contrastive loss functions. Simultaneously, a self-supervised decoder reconstructs the masked regions of the images, further enhancing the model's ability to learn spatial structures and semantic information. In the fine-tuning stage, the decoder is removed, leaving only the encoder, and an MLP classifier is introduced for few-shot classification training.

B. Data Augmentation Methods for Multimodality Using Mask Strategies

Mask-based data augmentation offers several significant benefits. By masking portions of the data, it generates multiple views for contrastive learning, which enhances the model's ability to learn robust and diverse representations. Additionally, the use of a decoder to reconstruct the obscured regions enforces the learning of spatial structures and semantic information, improving the model's understanding of the underlying data. This dual approach not only promotes better feature extraction and representation but also increases the model's resilience and discriminative power, leading to more accurate and reliable performance in various tasks.

We denote the hyperspectral data as $X_H \in \mathbb{R}^{W\times H\times C}$, where W and H represent the width and height of the HSI, respectively, and C is the number of spectral bands. For the target pixel to be classified, the corresponding $1 \times 1 \times C$ region in the hyperspectral data is designated as the spectral

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modality $\mathbf{X}_s \in \mathbb{R}^{1\times1\times C}$. Based on this pixel, a spatial patch of size $S \times S$ is extracted to obtain spatial modality information. To increase diversity in the spatial modality and to avoid interference caused by similar values in certain bands for different land cover types [37], a single band is randomly selected from the $S \times S$ neighborhood and is treated as the spatial modality $\mathbf{X}_a \in \mathbb{R}^{S\times S\times1}$. By increasing the number of iterations during training, coverage of all bands in the model is ensured. For the LiDAR modality, a region of size $S \times S$ is extracted from the LiDAR data at the same spatial position as $\mathbf{X}_s$, denoted as $\mathbf{X}_l \in \mathbb{R}^{S\times S\times1}$.

To enable the model to fully exploit high-level semantic features from both hyperspectral and LiDAR data, we apply masking to the spatial and LiDAR patches. This procedure increases data diversity and partially simulates real-world disturbances from factors such as illumination or fog, thereby improving the model's generalization performance. Meanwhile, a decoder reconstructs the masked portions, allowing the model to focus on local information in each sample and leverage the semantic similarity among neighboring regions in the HSI.

Specifically, we first apply convolution to the spatial patch of size $S \times S$ and divide it into $N = (S \times S/p^2)$ nonoverlapping patches, where p is the side length of each patch. Next, based on the hyperparameter ratio, we generate a masking index set

$$\mathcal{I} = \{a_1, a_2, \ldots, a_N \mid a_i \in \{0, 1\}\}$$

and apply masking to patches indexed by 1. To reduce the computational burden on the encoder and encourage deeper analysis of visible features [50], only the unmasked patches are retained to form the subset $\mathbf{X}_a^{\text{sub}} \in \mathbb{R}^{(N-n)\times C\times S\times S}$, which is then fed into the encoder (where n is the number of masked patches). The same procedure is applied to the LiDAR modality to obtain $\mathbf{X}_l^{\text{sub}}$. For the spectral modality, we randomly remove some bands at the same masking ratio to form $\mathbf{X}_s^{\text{sub}}$, and variations in sequence length do not adversely affect the training of the Transformer model, which is one of the key advantages that motivated us to choose Transformer as the baseline model.

According to related research [36], masking all three modalities simultaneously may hinder effective model fitting. Therefore, when the spatial and LiDAR modalities are masked, the spectral modality remains unmasked and vice versa, ensuring that at least one complete modality is available to guide the contrastive learning process.

C. Transformer-Based Encoder With KAN

Transformers excel at capturing long-range dependencies in data. However, when combined directly with MLP modules, they often lead to a significant increase in the number of parameters and computational overhead, which can easily cause overfitting under small-sample conditions. To address this issue, we introduce the KAN into the Transformer to replace the traditional MLP. By defining spline-based nonlinear mappings on the edges, KAN can significantly reduce the network's parameter count and computational cost while still maintaining strong feature representation capabilities, thereby better modeling complex spectral–spatial relationships and effectively mitigating the risk of overfitting.

The encoder is designed similar to the Vision Transformer, consisting of multiple stacked modules. Each module includes an MSA mechanism and a KAN structure. The feature update process for each module is illustrated in Fig. 1. The feature representation at the lth layer is calculated using the following equations:

$$\widetilde{\mathbf{f}}^{(l)} = \text{MSA}\left(\text{LN}\left(\mathbf{f}^{(l-1)}\right)\right) + \mathbf{f}^{(l-1)} \tag{1}$$

$$\mathbf{f}^{(l)} = \text{KAN}\left(\text{LN}\left(\widetilde{\mathbf{f}}^{(l)}\right)\right) + \widetilde{\mathbf{f}}^{(l)} \tag{2}$$

where LN denotes the layer normalization operation, MSA represents the multihead self-attention mechanism, and KAN refers to the Kolmogorov–Arnold network. Features are updated iteratively layer by layer to capture long-range dependencies in the input data.

To further explain the structure of the KAN, as illustrated in Fig. 1, the nodes in KAN perform only simple addition operations, while the nonlinear activation functions are defined on the edges. The network consists of L layers, and its computation can be expressed as follows:

$$\text{KAN}(x) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \cdots \circ \Phi_0)(x) \tag{3}$$

where Φ_l denotes the activation function matrix at the lth layer, with dimensions $n_{l+1} \times n_l$. Each matrix element, $\phi_{l,j,i}(\cdot)$, is a learnable nonlinear function defined as

$$\phi(x) = w(b(x) + \text{spline}(x)).$$

$b(x)$ is the base activation function, defined as the SiLU activation function

$$b(x) = \text{silu}(x) = \frac{x}{1 + e^{-x}}$$

and $\text{spline}(x)$ is a spline function, which can be expressed as

$$\text{spline}(x) = \sum_i c_i B_i(x)$$

where $B_i(x)$ represents the basis functions and c_i is the corresponding learnable parameter.

By defining learnable nonlinear functions on the edges, the KAN architecture achieves greater flexibility in modeling complex high-dimensional relationships. Compared to traditional neural network architectures, this design provides enhanced expressive power, making it particularly well-suited for tasks requiring precise modeling of nonlinear interactions.

D. Decoder Architecture for Multimodal Data Reconstruction

Although contrastive learning enforces similarity across modalities, it does not explicitly guarantee that local structures and contextual details are adequately modeled, especially when some patches or bands are masked. To address this gap, we incorporate a dedicated decoder to reconstruct the masked regions. This self-supervised objective compels the Transformer-based encoder to capture richer spatial-semantic information, thus enhancing the quality of learned embeddings.

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Feature vectors $f_a \in \mathbb{R}^{(N-n)\times C\times L}$, $f_l \in \mathbb{R}^{(N-n)\times C\times L}$, and $f_s \in \mathbb{R}^{(N-n)\times C\times 1}$ are extracted from the encoder, encapsulating the global contextual information of the samples. Subsequently, based on the indices I, a set of masked blocks $f_{a_m} \in \mathbb{R}^{n\times C\times L}$, $f_{l_m} \in \mathbb{R}^{n\times C\times L}$, and $f_{s_m} \in \mathbb{R}^{n\times C\times L}$ are generated for the positions originally covered by the mask. These masked blocks are shared and learnable vectors designed to fill in the masked regions.

The original feature vectors and the masked feature vectors are then reassembled into a sequence based on the patch structure of the original image and fed into the decoder. The structure of the Transformer layer is illustrated in Fig. 1. The decoder consists of N_d Transformer layers and two linear layers. Since the primary objective of the masking mechanism is to learn the local neighborhood information of the samples, the decoder is designed to be narrower than the encoder. Specifically, the first linear layer in the decoder reduces the dimensionality of the input feature vectors, while position encodings and class tokens are embedded into the sequence. These features are then processed through N_d Transformer layers.

Finally, the output of the decoder is projected into pixel space through the last linear layer, reconstructing the corresponding modal pixel: spatial modality f_{d_a}, spectral modality f_{d_s}, and LiDAR modality f_{d_l}. This process aims to efficiently reconstruct and integrate multimodal data.

E. Loss Function

To unify cross-modal alignment and self-supervised reconstruction, we adopt a dual-objective loss function comprising contrastive loss and reconstruction loss. This design addresses the critical challenge of simultaneously ensuring discriminative representation learning and capturing local details, thereby fostering a more robust and comprehensive feature space for subsequent few-shot classification.

During the pretraining phase, our model employs two principal loss functions. The first loss function is a contrastive learning loss aimed at facilitating the learning of discriminative features. In this process, the encoder extracts features f_a, f_l, and f_s from the input data, which are subsequently projected into a latent feature space through the KAN network, yielding the representations z_a, z_l, and z_s, respectively. To effectively capture the correlations between 2-D spatial information and the LiDAR modality, the representations z_l and z_s are concatenated and further fused using an additional KAN network, resulting in a complementary feature representation denoted as z_{la}.

In addition, to capture the shared high-level semantic features of the same sample across different modalities, we introduce four distinct contrastive learning loss functions. These loss functions are designed to constrain the feature learning process, thereby ensuring that the model extracts modality-invariant semantic information, which is critical for robust performance in downstream tasks

$$l_{as} = -\log \frac{\exp\left(\text{sim}\left(z_a^i, z_s^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_a^i, z_s^k\right)/\tau\right)} \tag{4}$$

$$l_{ls} = -\log \frac{\exp\left(\text{sim}\left(z_l^i, z_s^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_l^i, z_s^k\right)/\tau\right)} \tag{5}$$

$$l_{sl} = -\log \frac{\exp\left(\text{sim}\left(z_a^i, z_l^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_a^i, z_l^k\right)/\tau\right)} \tag{6}$$

$$l_{sla} = -\log \frac{\exp\left(\text{sim}\left(z_s^i, z_l a^i\right)/\tau\right)}{\sum_{k=1}^{N_b} I_{k\neq i} \exp\left(\text{sim}\left(z_s^i, z_l a^k\right)/\tau\right)}. \tag{7}$$

The second loss function is a reconstruction loss designed to guide the Transformer model toward a deeper understanding of the semantic relationships among adjacent regions within a sample. Specifically, this loss function focuses on the discrepancies between the reconstructed outputs for the masked regions and the corresponding portions of the original sample, quantified using mean squared error (mse) as the evaluation metric. This approach not only compels the model to accurately reconstruct local semantic features but also enhances its ability to capture and integrate contextual information in the masked areas

$$l_{\text{rec}_a} = \|y_a - f_{d_a}\|^2 \tag{8}$$

$$l_{\text{rec}_s} = \|y_s - f_{d_s}\|^2 \tag{9}$$

$$l_{\text{rec}_l} = \|y_l - f_{d_l}\|^2 \tag{10}$$

where $y_a \in \mathbb{R}^{(n)\times C\times L}$, $y_s \in \mathbb{R}^{(N-n)\times C\times 1}$, and $y_l \in \mathbb{R}^{n\times C\times L}$ is a sequence composed of masked patches.

By summing the contrastive and reconstruction loss functions, we obtain a composite loss that is used to update the parameters of both the encoder and the decoder. This combined loss ensures that the model simultaneously learns discriminative features through contrastive learning and effectively captures local semantic context through reconstruction, thereby enhancing overall performance.

III. Experiments

To comprehensively evaluate our proposed method, we conducted a series of experiments utilizing four multisensor datasets: Houston2013, MUUFL Gulfport, Trento, and Augsburg. Initially, we delineate comprehensive details regarding these datasets, subsequently offering an elucidation of our experimental setting. Extensive ablation experiments were performed to validate the effectiveness of each module. Finally, our classification results were compared with state-of-the-art methods from recent years to demonstrate the superior classification performance of our approach.

A. Dataset Description

1) Houston2013: The Houston2013 dataset encompasses imagery obtained via the ITRES CASI-1500 sensor, targeting the University of Houston campus and its adjacent urban environs. This sensor, operational across the spectral wavelength range of 0.38–1.05 μm, acquires data across 144 spectral bands. The dataset presents both hyperspectral and LiDAR images, each with dimensions of 349 $\times$ 1905 pixels. It comprises a total of 15 029 pixels to classify, segmented into 15 unique classes for the purpose of classification.

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS FOR HOUSTON2013, TRENTO, MUUFL, AND AUGSBURG DATASETS

No.	Houston2013 dataset			MUUFL dataset			Trento dataset			Augsburg dataset		
	Categories	Training	Test	Categories	Training	Test	Categories	Training	Test	Categories	Training	Test
C1	Healthy Grass	20	1241	Trees	20	23236	Apple Trees	20	4024	Forest	20	13497
C2	Stressed Grass	20	1244	Mostly Grass	20	4260	Buildings	20	2893	Residential Area	20	30319
C3	Synthetic Grass	20	687	Mixed Ground Surface	20	6872	Ground	20	469	Industrial Area	20	3841
C4	Trees	20	1234	Dirt and Sand	20	1816	Woods	20	9113	Low Plants	20	26847
C5	Soil	20	1232	Road	20	6677	Vineyard	20	10491	Allotment	20	565
C6	Water	20	315	Water	20	456	Roads	20	3164	Commercial Area	20	1635
C7	Residential	20	1258	Buildings Shadow	20	2223				Water	20	1520
C8	Commercial	20	1234	Buildings	20	6230						
C9	Road	20	1242	Sidewalk	20	1375						
C10	Highway	20	1217	Yello Curb	20	173						
C11	Railway	20	1225	Cloth Panels	20	259						
C12	Parking Lot1	20	1223									
C13	Parking Lot2	20	459									
C14	Tennis Court	20	418									
C15	Running Track	20	650									
Total		300	14879		220	53577		120	30154		140	78224

TABLE II
ABLATION STUDIES ON DIFFERENT DATASETS. ALL THE BEST ACCURACIES ARE IN BOLD

MLP	KAN	Projection	MAE	Houston2013	Muufl	Trento	Augsburg
✓	-	-	-	97.97	86.62	98.78	85.69
-	✓	-	-	98.08	86.95	99.4	86.89
-	✓	✓	-	98.14	87.47	99.62	88.29
-	✓	✓	✓	**98.63**	**88.78**	**99.66**	**90.13**

2) MUUFL: The MUUFL Gulfport dataset was acquired utilizing the Reflective Optics System Imaging Spectrometer (ROSIS) sensor stationed at the Gulf Park campus of the University of Southern Mississippi. This sensor delineates the picture in the wavelength interval from 0.375 to 1.05 μm, spanning 72 spectral bands. Owing to the presence of noise interference, eight of these bands were deemed unsuitable for analysis and were subsequently omitted, resulting in a final tally of 64 bands employed in the research endeavors. The dataset features both hyperspectral and LiDAR imagery, each with a resolution of 320 $\times$ 220 pixels. Encompassing a total of 55 337 pixels, the dataset is stratified into 11 distinct classes for classification purposes, including, but not limited to, trees, roads, and water.

3) Trento: The Trento dataset's HSI was procured via the AISA Eagle sensor, focusing on a rural locale situated to the south of Trento city. Concurrently, the LiDAR imagery was garnered utilizing the Optech ALTM3100EA sensor, characterized by a grid size of 1. The captured hyperspectral images are comprised of 63 spectral bands, with the spatial dimensions delineated as 600 $\times$ 166 pixels. Within this dataset, 30 214 pixels have been earmarked for classification endeavors, these pixels being systematically segregated into six distinct categories.

4) Augsburg: The Augsburg dataset was captured in the city of Augsburg. LiDAR data were obtained through the dlr-36 system, while the HSI were acquired using the DAS-EOC HySpex sensor. This sensor captures data in the wavelength range of 0.4–2.5 μm, encompassing 180 spectral bands. The dimensions of the HSIs are 332 $\times$ 485 pixels. A total of 78 294 pixels are utilized for classification in this dataset, covering seven different land cover categories.

The number of training samples and testing samples for each land cover category in the four datasets mentioned above is shown in Table I.

B. Experimental Setup

The model proposed in this article was implemented and evaluated on an Ubuntu 16.04 operating system with an NVIDIA GeForce RTX 4090 GPU, using the PyTorch framework. During the pretraining phase, the image patch size was set to 27, with a batch size of 256, and the model was trained for 100 iterations with a learning rate of $1e^{-3}$. The encoder consisted of four layers, each with eight attention heads, while the decoder included three layers. The masking ratio for the input patches was set to 0.4.

For the downstream classification tasks, the decoder was removed, retaining only the encoder. A KAN network was appended as the classifier. Each class was sampled with 20 instances, and the learning rates for the feature extractor and classifier were set to $1e^{-4}$ and $1e^{-3}$, respectively. The batch size was adjusted to 64, and training was conducted for 100 iterations.

C. Evaluation Metrics

To quantitatively assess the classification performance of the proposed model, three widely used evaluation metrics are adopted in this study: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. These metrics offer a comprehensive evaluation from different perspectives, enabling objective comparison across models and datasets.

1) *OA:* It is defined as the ratio of the number of correctly classified samples to the total number of samples in the test set. It reflects the global classification performance of the model across all classes.

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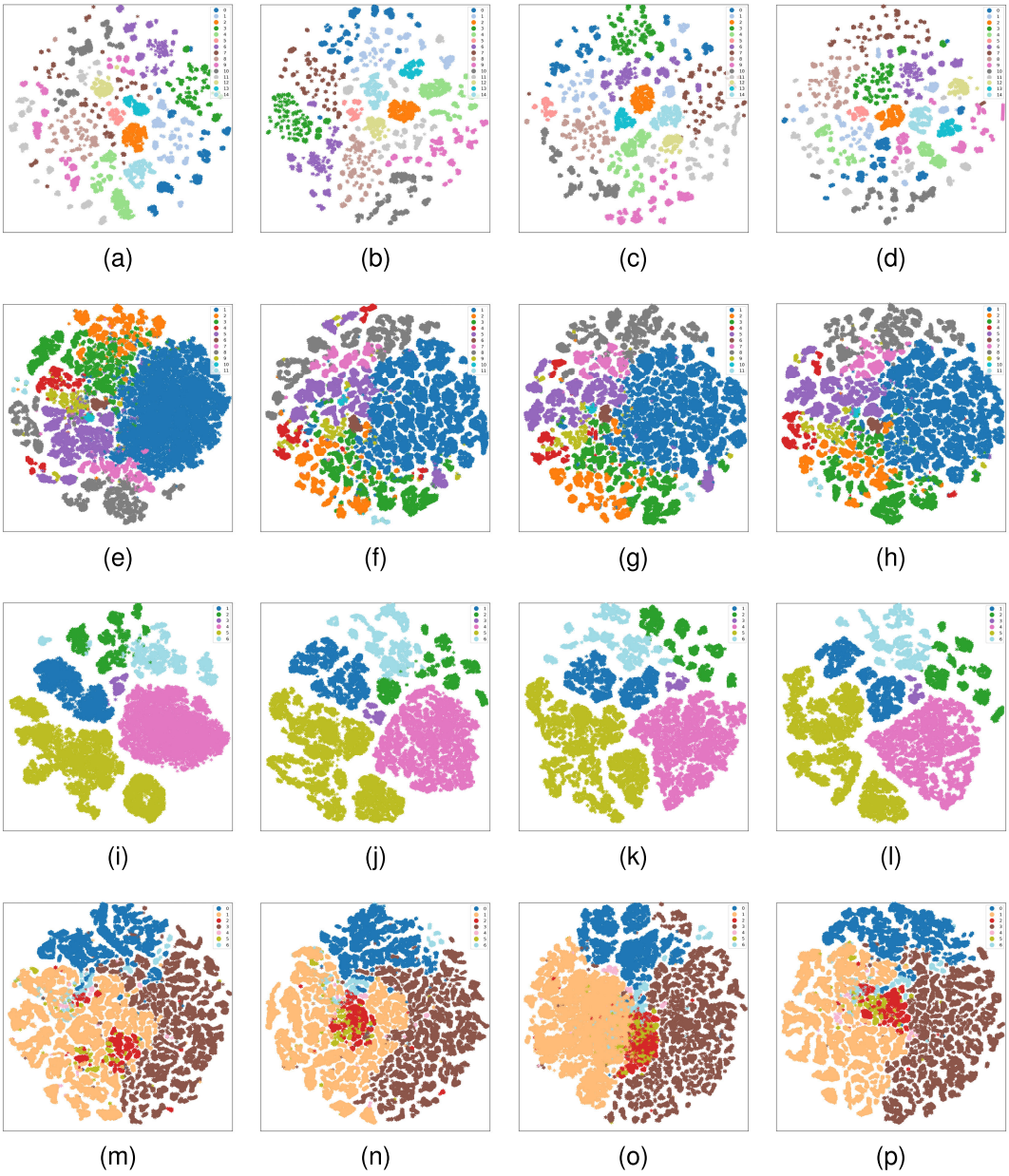


Fig. 3. Feature visualization of encoders for HSI and LiDAR classification on four datasets. (a)–(d) Results on the Houston2013 dataset. (e)–(h) Results on the MUUFL dataset. (i)–(l) Results on the Trento dataset. (m)–(p) Results on the Ausburg dataset. The first column represents the use of the original Transformer model, the second column represents replacing the MLP with the KAN network, the third column represents using a projection layer to fuse spatial and LiDAR modalities, and the fourth column represents using a decoder to reconstruct the masked patches.

2) *AA:* It is the mean of the individual classification accuracies of each class. It emphasizes balanced performance and is particularly useful when dealing with class-imbalanced datasets.
3) *Kappa Coefficient:* It measures the agreement between the predicted classification and the ground truth while taking into account the agreement occurring by chance. It provides a more robust evaluation, especially in the presence of skewed class distributions.
4) *Floating-Point Operations (FLOPs):* They refer to the total number of floating-point operations required for a single forward pass through the model. This metric provides a hardware-agnostic estimation of the computational complexity of different architectures. In remote sensing tasks, where input data often have high spatial resolution and spectral dimensionality, FLOPs serve as an important reference for understanding the theoretical computational demand. Although FLOPs do not directly correspond to runtime performance—which also depends on hardware characteristics such as memory bandwidth and parallelism—they remain a standard measure to compare the efficiency of model designs. In this work, we report FLOPs to support future deployment decisions in resource-constrained environments.
5) *Parameter:* The parameter count refers to the total number of learnable weights in the model. It directly affects the storage requirements and memory consumption during both training and inference. In practical applications such as edge computing or low-power remote sensing platforms, a lower parameter count generally

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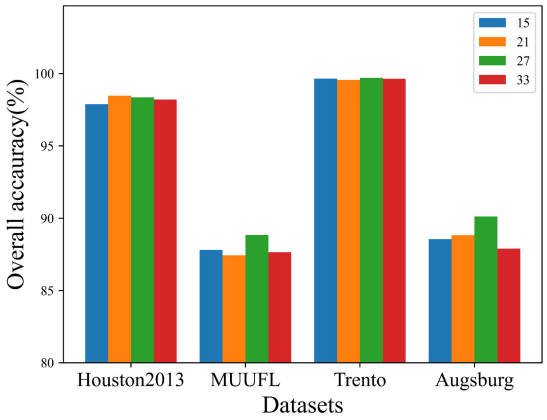


Fig. 4. Impact of different feature block size selections on classification accuracy.

favors efficient deployment. However, large models often offer stronger expressive capacity and convergence stability, especially in data-scarce scenarios. Therefore, we include parameter statistics alongside classification metrics to facilitate a comprehensive evaluation of the tradeoff between model complexity and performance. It is important to clarify that our method does not claim reduced computational costs as a core contribution but instead presents these figures for transparency and fair comparison.

These three metrics together ensure a thorough and balanced evaluation of the proposed model's effectiveness under various conditions.

D. Ablation Study

To comprehensively validate the effectiveness of each module, we designed a series of ablation experiments to evaluate the impact of different module combinations on classification accuracy. The experimental results are shown in Table II. The experiments include four configurations: the first column demonstrates the case where only the Transformer encoder is used with contrastive learning for pretraining, the second column shows the configuration where the MLP layer is replaced with the KAN module, the third column incorporates an additional perceptron module to fuse spatial and LiDAR modalities, and the last column utilizes a decoder to reconstruct the three modalities.

Among these configurations, the KAN module improves the classification accuracy with fewer parameters and the same amount of training data, particularly achieving a 1.2% increase in accuracy on the Augsburg dataset. The fusion of spatial and LiDAR modalities effectively exploits the complementary information between the two modalities. Specifically, for the Houston dataset, due to the scattered class distribution and a large amount of background classes, the fusion of spatial and LiDAR information might introduce additional noise, resulting in only a 0.06% improvement in accuracy; however, for the Muffl and Augsburg datasets, which feature complex and dense spatial distributions, classification accuracy improved by 0.52% and 1.4%, respectively. Finally, by using the Transformer decoder to reconstruct the masked patch information from the three modalities, inductive bias was introduced into the model. This enabled the Transformer to better focus on the neighboring information of the target objects, fully leveraging the semantic similarity of adjacent areas in HSIs. This resulted in accuracy improvements of 0.49%, 1.31%, and 1.84% on the Houston, Muffl, and Augsburg datasets, respectively. It is worth noting that for the Trento dataset, which has a relatively balanced and homogeneous class distribution, the improvement was minimal, at only 0.04.

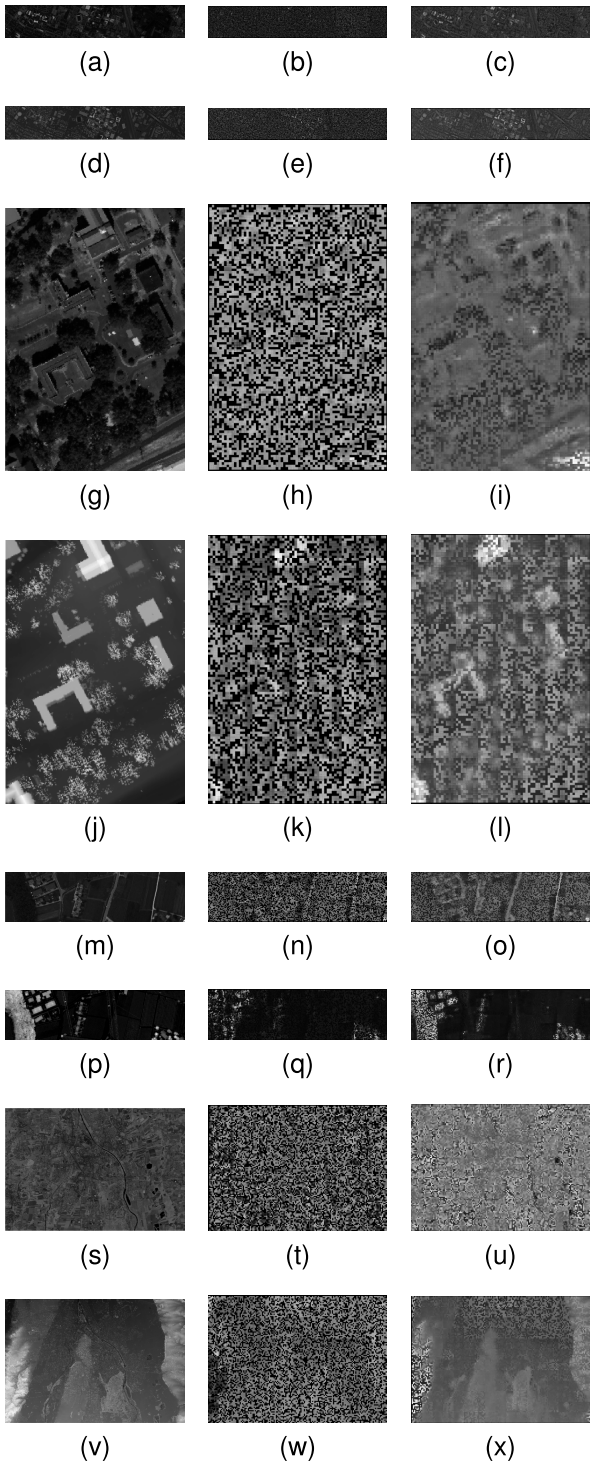


Fig. 5. Reconstruction examples obtained by our method. Each row corresponds to a different dataset: Houston2013, MUUFL, Trento, and Augsburg, respectively. The first column shows the original grayscale images of the two modalities, the second column shows the masked images, and the third column shows the reconstructed images.

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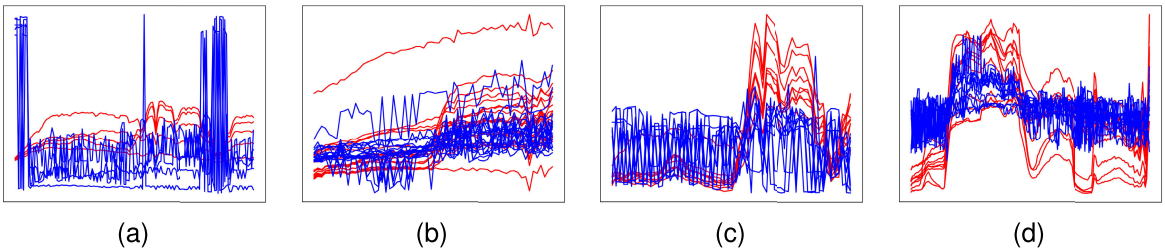


Fig. 6. Reconstruction spectrum obtained by our method on four different datasets. Blue represents the reconstructed spectrum, while red represents the original spectrum. Each subfigure (a)–(d) corresponds to the Houston2013, MUUFL, Trento, and Augsburg datasets, respectively.

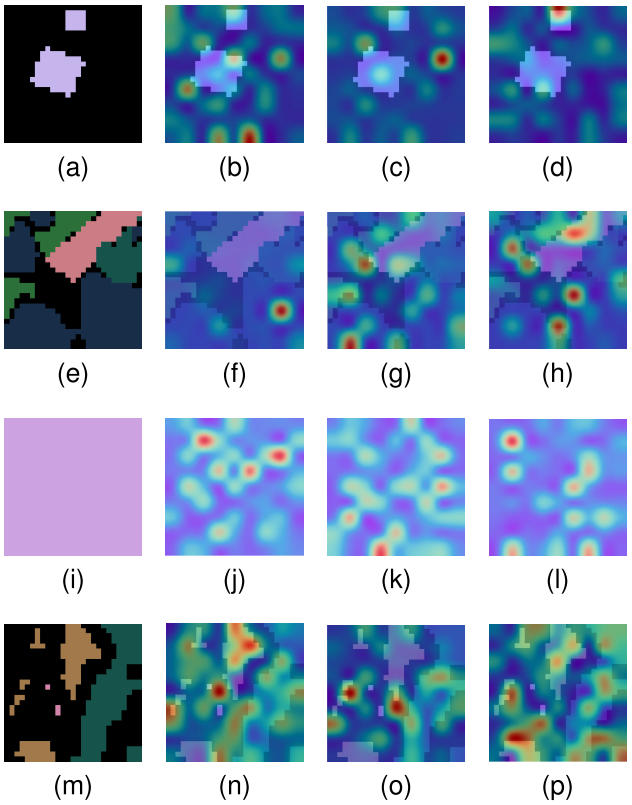


Fig. 7. Attention maps obtained by our method. (a), (e), (i), and (m) Ground truth. (b), (c), (f), (g), (j), (k), (n), and (o) Two randomly selected heads from the last layer of the image encoder. (d), (h), (l), and (p) One randomly selected head from the last layer of the LiDAR encoder.

Additionally, to further demonstrate the effectiveness of our method in feature extraction, we used t-distributed stochastic neighbor embedding (t-SNE) to visualize the features extracted under different module configurations. As shown in Fig. 3, after fusing spatial and LiDAR information, the clustering of data points improved significantly, with tighter clusters and clearer boundaries between classes. Moreover, the introduction of reconstruction loss emphasized the internal information, further enhancing the structural consistency within clusters and yielding clearer intraclass relationships.

In summary, our approach, through well-designed modules, not only achieved notable improvements in classification accuracy across multiple datasets but also demonstrated superior performance in feature extraction and visualization.

E. Computational Cost of the Model

This section provides a brief analysis of the computational cost of the proposed method, focusing on FLOPs and the number of model parameters, as shown in Table III. To better capture global features from images, we adopt a Transformer as the baseline model, which inevitably leads to a higher parameter count compared to other methods. However, it is noteworthy that the model based on KAN not only achieves higher classification accuracy than its MLP-based counterpart but also significantly reduces the number of parameters—from 3.51 to 3.11 M, a reduction of approximately 11%—with a slight decrease in FLOPs (from 375.05 to 374.65 M).

These results suggest that KAN is capable of learning more expressive and compact feature representations, thereby achieving improved performance with fewer parameters and marginally lower computational cost. Nonetheless, we argue that computational efficiency is not the central competitive advantage of our study. The primary contribution lies in exploring the feasibility of using Transformers within a contrastive learning framework for hyperspectral and LiDAR data fusion and in addressing the lack of inductive bias in Transformer models through a decoder mechanism. These findings have been thoroughly validated by extensive experiments.

F. Analysis of Patch Size

Given the significant influence of patch size on the model's ability to extract semantic features, an ablation study with varying patch sizes was conducted, and the results are presented in Fig. 4. However, as the spatial extent of the patch increases, the model may be affected by external factors such as variations in sensor angles, different environmental conditions (e.g., humidity and weather), and other influences. These factors can lead to the "same object, different spectrum" phenomenon for distant features. Additionally, larger patches may introduce more irrelevant information, such as pixels from unrelated objects, which can degrade classification accuracy.

Specifically, according to Fig. 4 for the Muffl, Trento, and Augsburg datasets, the classification performance is optimal when the patch size is set to 27. This can be attributed to the relatively concentrated class distributions in these datasets, where a moderately larger patch effectively captures more semantic information. In contrast, for the Houston dataset, the class distribution is more scattered and point-like. In this case, larger patches tend to introduce more unrelated information, making a patch size of 15 more effective for classification.

Based on this analysis, subsequent experiments will use the optimal patch size for each dataset to ensure the best classification performance.

G. Analysis of Reconstruction Examples

Fig. 5 shows the reconstructed images of four datasets. To better visualize the reconstruction effects, we fixed a single

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TABLE III
COMPUTATIONAL COST OF THE PROPOSED METHOD

Dataset	Complexity	EndNet	S2EnNet	Fusion-HCT	DSHFNet	CMCL	CACL	TMCL	Ours-MLP	Ours-KAN
Houston2013	FLOPS	92.26K	13.27M	7.93M	55.30M	96.64M	31.70M	146.16M	375.05M	374.65M
	Parameters	91.97K	270.80K	429.84K	332.46K	13.38M	295.68K	26.47M	3.51M	3.11M

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE HOUSTON2013 DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	TMCL	Ours
1	95.55	**98.30**	97.32	95.28	85.30	96.51	96.63	97.92
2	98.79	**99.52**	98.87	91.00	83.06	99.03	98.91	95.61
3	99.85	99.12	99.41	98.52	97.78	97.64	100	**100**
4	93.73	93.82	98.94	99.59	99.02	97.39	99.45	**100**
5	95.76	100	100	98.85	97.38	100	99.68	**99.92**
6	95.81	99.03	100	85.57	79.02	92.46	100	**100**
7	78.13	84.68	92.47	91.99	94.95	94.31	97.72	**99.21**
8	88.93	95.04	95.26	81.29	96.32	93.55	95.24	**98.23**
9	71.14	78.50	85.23	76.78	72.32	86.44	94.12	**98.08**
10	68.56	77.89	86.25	82.02	82.35	93.95	90.47	**98.61**
11	83.85	97.70	97.20	96.63	90.62	97.70	99.63	**100**
12	85.71	**95.98**	82.28	93.40	69.74	89.78	93.77	95.94
13	53.96	95.81	98.22	97.55	98.00	**99.78**	98.86	98.29
14	97.58	100	100	100	100	100	100	**100**
15	99.07	100	100	96.72	98.73	99.84	100	**100**
OA	86.78	93.28	94.43	91.80	88.55	95.45	97.18	**98.58**
AA	87.10	94.36	95.43	92.55	89.51	95.89	97.65	**98.78**
Kappa	85.70	92.73	93.97	91.13	87.63	95.08	96.96	**98.46**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE MUUFL DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	TMCL	Ours
1	69.82	77.63	84.92	74.38	**89.34**	88.12	86.37	87.60
2	82.78	82.68	71.92	79.72	79.20	68.96	71.99	**88.95**
3	64.60	58.47	57.31	73.03	64.98	45.23	82.44	**85.48**
4	70.10	70.82	86.12	92.97	77.41	73.53	**95.07**	94.58
5	89.08	83.46	76.12	**90.49**	82.36	88.20	84.02	86.93
6	99.10	88.34	100	100	99.78	100	97.37	**100**
7	91.28	92.95	82.47	94.49	92.36	91.87	89.35	**97.44**
8	87.09	88.73	88.86	93.78	92.03	92.56	92.62	**94.98**
9	64.76	64.98	62.95	43.44	65.49	57.88	31.84	**79.42**
10	95.09	87.73	83.59	0	74.85	60.74	87.44	**97.81**
11	96.79	97.99	97.99	99.19	94.78	99.20	87.44	**100**
OA	75.81	77.89	79.12	79.67	84.03	80.58	84.61	**88.85**
AA	82.77	81.25	80.30	76.50	82.96	78.75	77.35	**92.10**
Kappa	69.98	72.33	73.37	74.54	79.38	75.02	79.94	**85.66**

TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE TRENTO DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	TMCL	Ours
1	73.17	99.18	99.02	99.85	99.75	98.48	99.55	**99.72**
2	96.50	96.12	97.20	97.81	98.43	99.17	96.35	**98.83**
3	92.59	100	97.77	99.78	96.51	100	98.64	**100**
4	95.55	99.62	99.87	99.59	**100**	100	100	99.98
5	69.10	99.83	99.19	99.74	99.53	99.91	98.90	**100**
6	89.79	95.12	92.07	97.90	91.60	91.37	98.04	**98.61**
OA	82.79	98.83	98.42	99.34	98.72	98.78	99.04	**99.70**
AA	86.12	98.31	97.52	99.12	97.64	98.16	98.58	**99.52**
Kappa	77.58	98.44	97.89	99.11	98.29	98.38	98.69	**99.60**

band during the data preprocessing phase instead of randomly selecting one. For each dataset, in the first row, the first column shows the grayscale image of a specific hyperspectral band, the second column shows the image after masking, and the third column shows the reconstructed image. In the second row, the first column displays the grayscale LiDAR image, the second column shows the LiDAR image after masking, and the third column shows the reconstructed LiDAR image.

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Fig. 6 presents the spectral reconstruction results of the four datasets. The red curve represents the original spectrum, while the blue curve represents the reconstructed spectrum. From the reconstructed images, it is clear that although some detailed information is lost, the reconstructed images still retain much of the contextual information. Specifically, the Houston dataset, which contains a large proportion of background classes, exhibits the best reconstruction quality with a good recovery of details. In contrast, the Augsburg dataset has a more complex spatial distribution, as seen in the original grayscale image, resulting in poorer reconstruction. However, even in this case, key features like the road on the left are successfully reconstructed based on contextual information. While the spectral details are somewhat degraded, the reconstructed spectral curves accurately reflect the overall trends, showing a precise spectral reconstruction.

These results demonstrate that although some information is inevitably lost during reconstruction, especially at the detail level, the recovery of contextual information and overall spectral trends remains satisfactory. These findings provide valuable insights for future research on hyperspectral and LiDAR data reconstruction.

H. Analysis of Attention Maps

Fig. 7 visualizes attention maps across different datasets. Specifically, the first column shows the ground truth for each dataset; the second and third columns display the attention maps from the last layer of the spatial modality encoder, with two randomly selected heads; and the fourth column shows the attention map from the last layer of the LiDAR encoder, with one selected head.

From the visualization results of the Houston dataset, it can be seen that the attention maps of each head primarily focus on the upper edges of similar objects, which helps in the classification of central objects. This observation indicates that the attention mechanism effectively captures long-range dependencies.

In the Muffl dataset, the attention maps emphasize the same objects in different regions. These objects are located in three distinct areas, and the model assigns corresponding attention to them. This further demonstrates that the attention mechanism can effectively focus on similar targets in different spatial locations.

Moreover, for the Augsburg dataset, the attention maps from the spatial modality and LiDAR modality focus on different areas. The complementarity between the two modalities allows the model to capture more accurate target information in the scene, thereby enhancing the accuracy of scene understanding.

In conclusion, the attention maps from different datasets showcase the strong capabilities of the attention mechanism in handling multimodal data, particularly in constructing long-range dependencies and providing complementary information, which greatly enhances the model's recognition and classification abilities.

I. Comparison of Classification Results

We compared the proposed method with several typical multimodal classification methods to validate the superiority of our model. The following is a brief description of these methods.

1) *EndNet [51]:* This method uses a deep encoder–decoder structure to explore the latent associations between multimodal data, enabling effective information fusion.
2) *S2ENet [52]:* It enhances cross-modal information interaction by introducing a spectral enhancement module, improving the overall performance of multimodal learning.
3) *Fusion-HCT [53]:* This method designs a hierarchical convolution and Transformer framework to achieve a collaborative representation of multimodal data, fully exploiting the potential of different modalities.
4) *DSHFNet [54]:* It divides samples into multiple scales and integrates information from different scales based on thresholds, overcoming the limitations of single-scale feature extraction and improving classification performance.
5) *CACL [55]:* Based on generative adversarial networks (GANs), CACL designs a multimodal dual-generator-discriminator structure and uses a voting mechanism to fuse features at different levels, enhancing the classification accuracy of multimodal data.
6) *CMCL [56]:* It designs a multiview contrastive learning framework and introduces additional fine-tuning losses to optimize feature transfer, improving the robustness of cross-modal learning.
7) *TMCL [49]:* This method preprocesses hyperspectral data into spatial and spectral modalities, designing a triplet contrastive learning framework to strengthen the representational ability of multimodal learning.

Tables IV–VII present a comparison of our method with seven other methods in terms of classification accuracy on four datasets, evaluated using OA, AA, and Kappa coefficients for each category. The best results among all methods are highlighted in bold.

For the Houston dataset, the sample distribution is point-like, and the spatial extent of each region is relatively small. Therefore, the DSHFNet model is unable to fully leverage large-scale information. In this context, Fusion-HCT, with its CTA fusion encoder, effectively integrates HSI and LiDAR data, significantly improving classification accuracy by 2.63% compared to DSHFNet. However, CMCL, with its relatively simple feature extractor structure, is unable to effectively extract complex features when dealing with large amounts of data, resulting in lower accuracy for this dataset. Our method achieved the highest accuracy in 11 out of 15 classes. Fig. 8 shows the classification results of various methods on the Houston2013 dataset. Even in areas with fragmented distributions on the right side, our method performs well, demonstrating that the image reconstruction technique helps the model better utilize local information, thereby improving classification accuracy in point-like regions.

The Muffl dataset features a relatively dense sample distribution with irregular spatial shapes, and there are often transitional zones between different classes. Using regular spatial patches to extract features may introduce irrelevant

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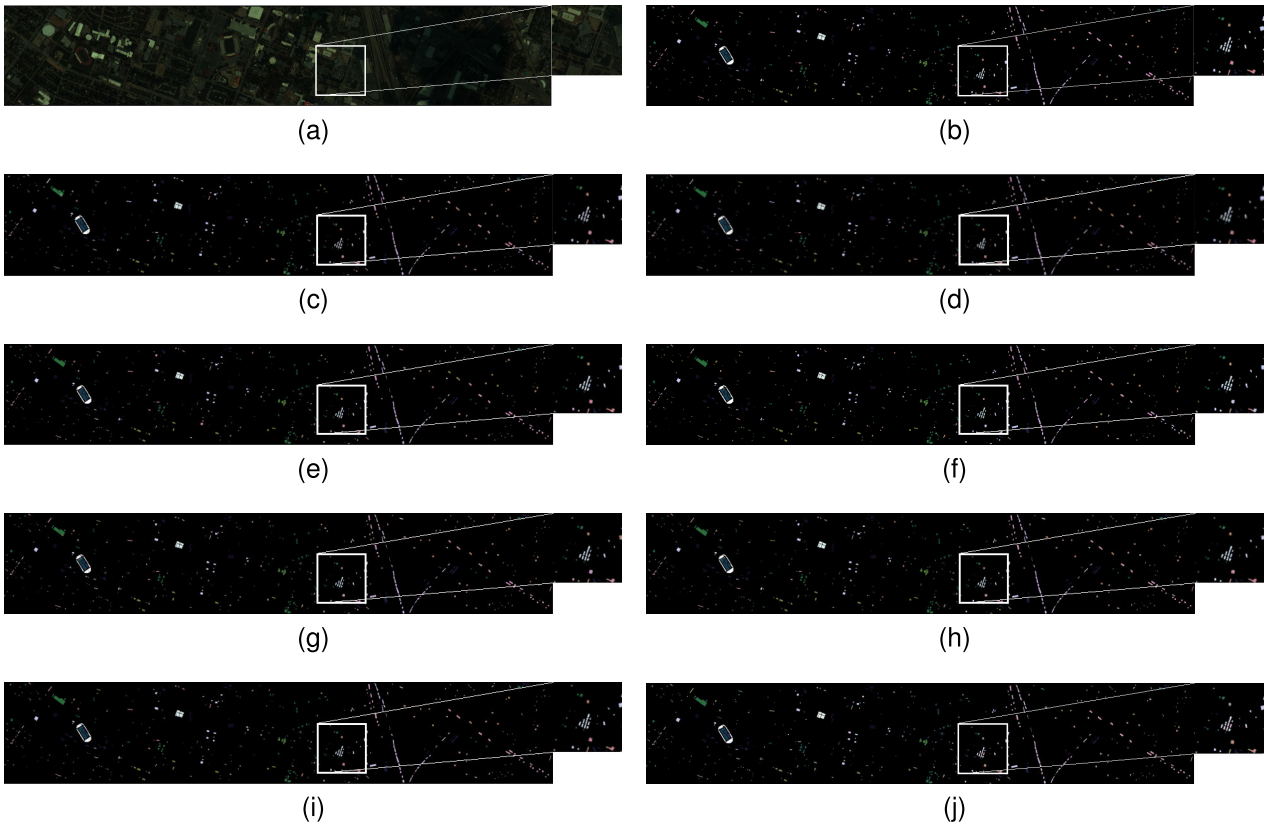


Fig. 8. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color, (b) ground truth, (c) EndNet (86.78%), (d) S2ENet (93.28%), (e) Fusion_HCT (94.43%), (f) DSHFNet (91.80%), (g) CMCL (88.55%), (h) CACL (95.45%), (i) TMCL (97.18%), and (j) proposed method (98.35%).

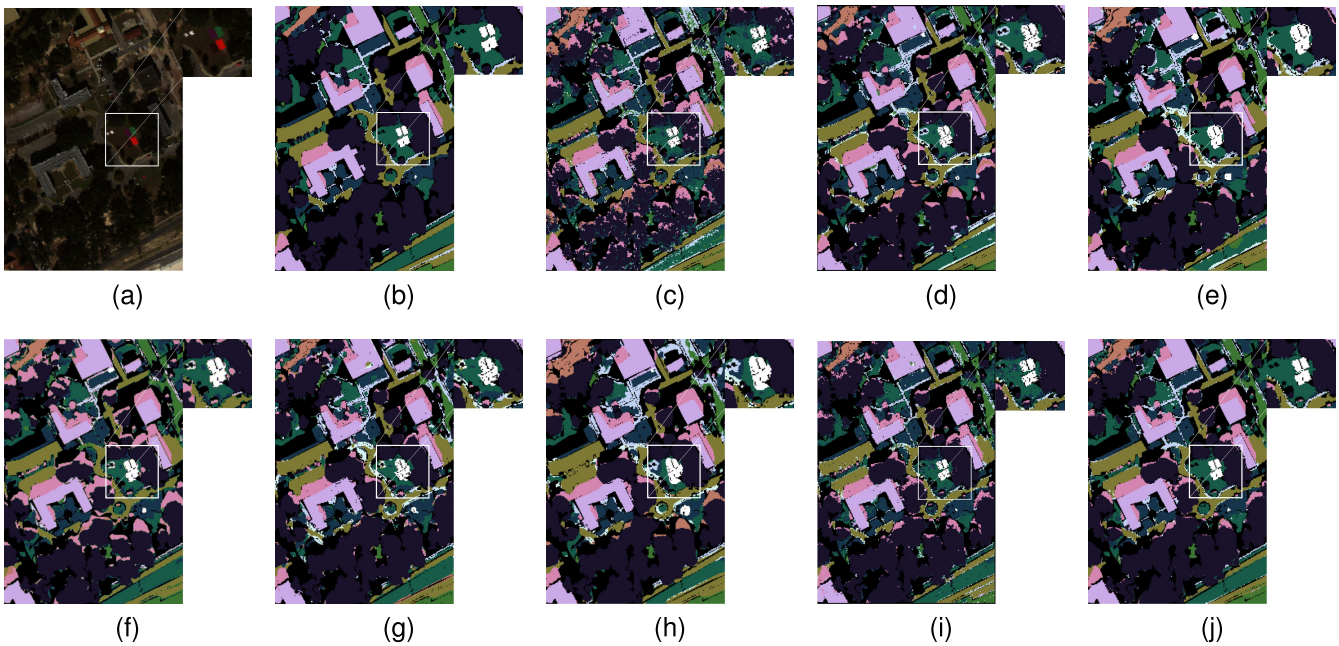


Fig. 9. Classification maps of each algorithm on the Mufflg dataset. (a) Pseudo-color, (b) ground truth, (c) EndNet (75.81%), (d) S2ENet (77.89%), (e) Fusion_HCT (79.12%), (f) DSHFNet (79.67%), (g) CMCL (84.03%), (h) CACL (80.58%), (i) NNCNet (84.61%), and (j) proposed method (88.85%).

interference from nontarget land cover, negatively affecting classification accuracy. DSHFNet addresses this issue through a multiscale approach, achieving the highest accuracy among supervised methods. TMCL, employing a tri-modal contrastive learning framework, reduces the impact of nontarget pixels in spatial blocks by utilizing the mutual information between spectral and spatial modalities. Our method, by effectively combining spatial modalities with LiDAR data, captures the 3-D representation of the Muffl dataset, achieving the best accuracy in 7 out of 11 classes. Compared to TMCL, our method improves OA, AA, and Kappa coefficients by 4.24%, 14.75%, and 5.72%, respectively. Fig. 9 illustrates the classifi-

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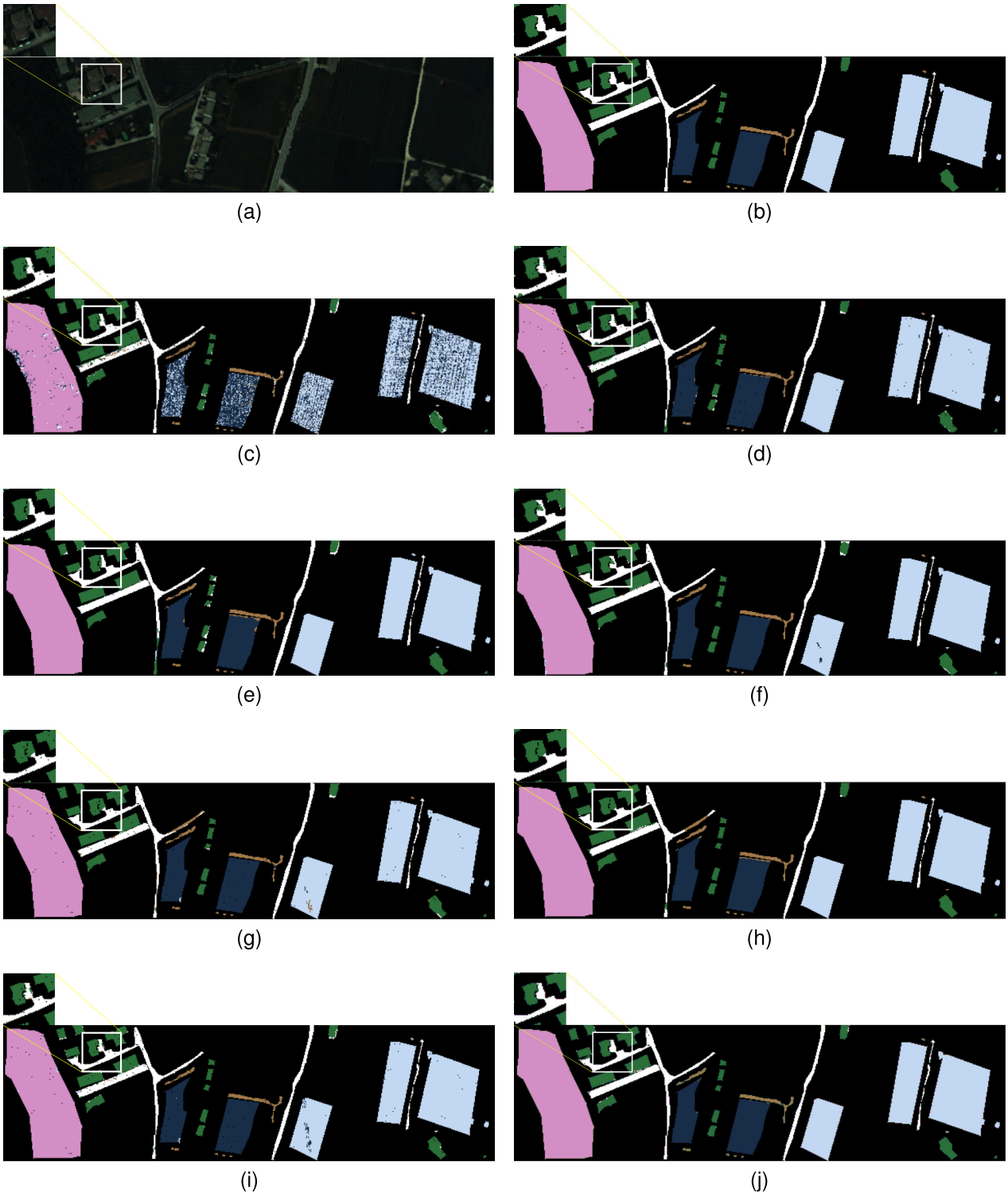


Fig. 10. Classification maps of each algorithm on the Trento dataset. (a) Pseudo-color, (b) ground truth, (c) EndNet (82.79%), (d) S2ENet (98.83%), (e) Fusion_HCT (98.42%), (f) DSHFNet (99.34%), (g) CMCL (98.72%), (h) CACL (98.78%), (i) NNCNet (99.04%), and (j) proposed method (99.70%).

cation results of different methods on the Muffl dataset, where it can be observed that our method still performs well in areas with highly irregular distributions, especially in the top-right corner.

The Trento dataset has a complete spatial distribution with high homogeneity and low intraclass heterogeneity. In this context, EndNet, which uses only spectral information, significantly underperforms compared to other methods, especially falling behind S2ENet by 16.04%. In contrast to TMCL, which uses small convolutional patches for classification, our method leverages a Transformer architecture to establish global dependencies, achieving a 0.66% higher accuracy than TMCL. Our method achieves optimal accuracy in four out of six classes. Fig. 10 shows the classification results of different methods on the Trento dataset, where our method performs well across large spatial blocks, with nearly all areas classified correctly.

The Augsburg dataset has a complex and highly variable spatial distribution with a mixed pattern of regions, as well as

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TABLE VII
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THE AUGSBURG DATASET

Class No.	EndNet	S2ENet	Fusion-HCT	DSHFNet	CMCL	CACL	TMCL	Ours
1	95.33	92.47	95.65	**99.36**	94.72	92.51	98.56	98.76
2	42.20	83.38	74.94	54.41	79.45	95.39	83.84	**87.08**
3	43.25	39.36	60.30	35.00	54.14	46.05	70.39	**72.83**
4	64.47	80.97	91.42	92.02	90.84	78.88	**97.22**	94.01
5	85.59	90.45	96.21	92.61	87.21	67.57	87.39	**98.61**
6	44.55	66.89	60.49	69.97	65.91	**70.77**	27.60	59.15
7	66.16	72.45	70.86	59.54	71.19	72.05	**81.36**	80.00
OA	59.89	81.46	83.23	74.83	84.37	85.64	89.52	**90.13**
AA	63.08	75.14	78.55	71.85	77.64	79.84	78.05	**84.35**
Kappa	49.37	74.81	77.30	78.67	66.90	74.47	82.23	**86.29**

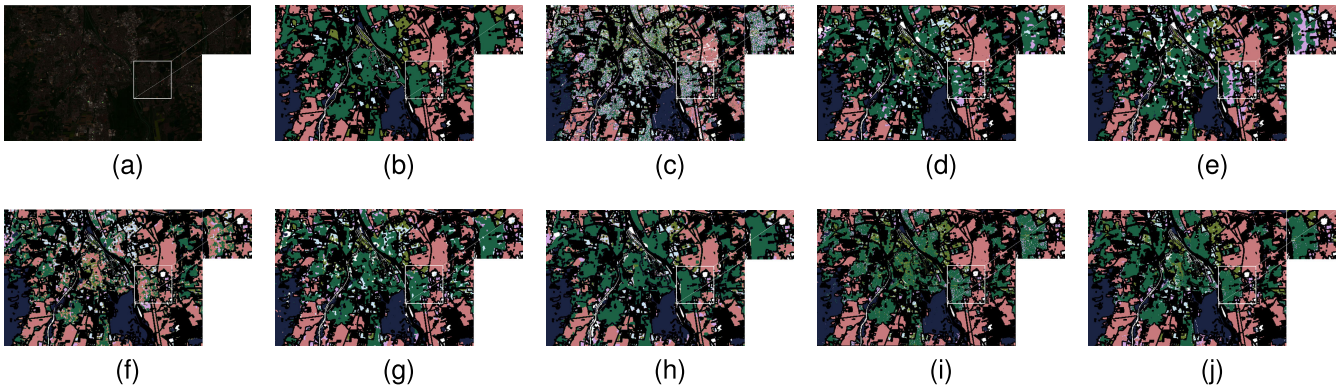


Fig. 11. Classification maps of each algorithm on the Pavia University dataset. (a) Pseudo-color, (b) ground truth, (c) EndNet (59.89%), (d) S2ENet (81.46%), (e) Fusion_HCT (83.23%), (f) DSHFNet (74.83%), (g) CMCL (84.37%), (h) CACL (85.64%), (i) NNCNet (89.52%), and (j) proposed method (90.13%).

many small-category areas. In this case, DSHFNet's multiscale advantage is no longer as prominent, leading to a 6.63% reduction in accuracy compared to S2ENet, which employs a spectral enhancement module. TMCL, by utilizing tri-modal mutual information learning and making full use of unlabeled samples, achieves a 5.15% higher accuracy compared to CMCL (which uses dual modalities). Our method, by reconstructing patches to mine neighborhood information from samples, performs exceptionally well in improving classification accuracy. Compared to TMCL, our method achieves a 0.61% improvement in accuracy. As shown in Fig. 11, our method effectively reduces block-based misclassifications, particularly in the Augsburg dataset, where the spectral redundancy in hyperspectral data leads to misclassifications, mostly involving point-like errors.

IV. CONCLUSION

In this article, we propose a novel network architecture for multimodal contrastive learning in remote sensing, addressing key challenges such as high-dimensional feature fusion, insufficient modality interaction, and the limitations of traditional neural networks. We visualize attention maps to demonstrate the model's strong ability to capture long-range dependencies and use image reconstruction tasks to validate the decoder module's effectiveness in fully utilizing the neighborhood information of HSIs.

Extensive experiments on four public datasets validate the superiority of our approach in terms of classification accuracy, confirming its efficacy in remote sensing multimodal learning. Despite introducing the decoder module to enhance inductive bias, the model still faces challenges due to the irregular shapes of many ground objects, which can lead to interference from nontarget objects. Therefore, future research will focus on leveraging GNNs to address the issues posed by the irregularity of ground object shapes, further improving the model's performance and robustness in complex scenarios.

REFERENCES

[1] Y. E. García-Vera, A. Polochè-Arango, C. A. Mendivelso-Fajardo, and F. J. Gutiérrez-Bernal, "Hyperspectral image analysis and machine learning techniques for crop disease detection and identification: A review," *Sustainability*, vol. 16, no. 14, p. 6064, Jul. 2024.

[2] D. Li, J. Wu, J. Zhao, H. Xu, and L. Bian, "SpectraTrack: Megapixel, hundred-fps, and thousand-channel hyperspectral imaging," *Nature Commun.*, vol. 15, no. 1, p. 9459, Nov. 2024.

[3] S. N. Khonina, N. L. Kazanskiy, I. V. Oseledets, A. V. Nikonorov, and M. A. Butt, "Synergy between artificial intelligence and hyperspectral imagining—A review," *Technologies*, vol. 12, no. 9, p. 163, Sep. 2024.

[4] D. Lee, M. Jung, W. Yang, and A. Kim, "LiDAR odometry survey: Recent advancements and remaining challenges," *Intell. Service Robot.*, vol. 17, no. 2, pp. 95–118, Mar. 2024.

[5] S. M. Farhan, J. Yin, Z. Chen, and M. S. Memon, "A comprehensive review of LiDAR applications in crop management for precision agriculture," *Sensors*, vol. 24, no. 16, p. 5409, Aug. 2024.

[6] T. Tao et al., "LiDAR-NeRF: Novel LiDAR view synthesis via neural radiance fields," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 390–398.

[7] N. Farmonov, M. Esmaeili, D. Abbasi-Moghadam, A. Sharifi, K. Amankulova, and L. Mucsi, "HypsLiDNet: 3-D–2-D CNN model and spatial–spectral morphological attention for crop classification with DESIS and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11969–11996, 2024.

[8] Q. Feng, D. Zhu, J. Yang, and B. Li, "Multisource hyperspectral and LiDAR data fusion for urban land-use mapping based on a modified two-branch convolutional neural network," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 1, p. 28, Jan. 2019.

[9] B. Ye, S. Tian, Q. Cheng, and Y. Ge, "Application of lithological mapping based on advanced hyperspectral imager (AHSI) imagery onboard GaoFen-5 (GF-5) satellite," *Remote Sens.*, vol. 12, no. 23, p. 3990, Dec. 2020.

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[10] A. Wang et al., "Potential of hyperspectral LiDAR in individual tree segmentation: A comparative study with multispectral LiDAR," *Urban Forestry Urban Greening*, vol. 104, Feb. 2025, Art. no. 128658.

[11] S. Hajaj, A. El Harti, A. B. Pour, A. Jellouli, Z. Adiri, and M. Hashim, "A review on hyperspectral imagery application for lithological mapping and mineral prospecting: Machine learning techniques and future prospects," *Remote Sens. Applications: Soc. Environ.*, vol. 35, Aug. 2024, Art. no. 101218.

[12] T. Ma, W. Huang, C. Wang, S. Ge, and Y. Liu, "Hyperspectral image classification based on extrema morphological profiles and total variation," *J. Phys., Conf. Ser.*, vol. 2258, no. 1, Apr. 2022, Art. no. 012036.

[13] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[14] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-spatial Inf. Sci.*, vol. 26, no. 3, pp. 262–288, Jul. 2023.

[15] M. Yasir, W. Jianhua, L. Shanwei, H. Sheng, X. Mingming, and M. Hossain, "Coupling of deep learning and remote sensing: A comprehensive systematic literature review," *Int. J. Remote Sens.*, vol. 44, no. 1, pp. 157–193, Jan. 2023.

[16] J. Huang, Y. Zhang, F. Yang, and L. Chai, "Attention-guided fusion and classification for hyperspectral and LiDAR data," *Remote Sens.*, vol. 16, no. 1, p. 94, Dec. 2023.

[17] D. Yao et al., "Deep hybrid: Multi-graph neural network collaboration for hyperspectral image classification," *Defence Technol.*, vol. 23, pp. 164–176, May 2023.

[18] M. Wang, Y. Sun, J. Xiang, R. Sun, and Y. Zhong, "Joint classification of hyperspectral and LiDAR data based on adaptive gating mechanism and learnable transformer," *Remote Sens.*, vol. 16, no. 6, p. 1080, Mar. 2024.

[19] W. Yu, L. Gao, H. Huang, Y. Shen, and G. Shen, "HI^{2}D^{2}FNet: Hyperspectral intrinsic image decomposition guided data fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521715.

[20] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[21] S. Tian, L. Li, W. Li, H. Ran, X. Ning, and P. Tiwari, "A survey on few-shot class-incremental learning," *Neural Netw.*, vol. 169, pp. 307–324, Jan. 2024.

[22] H. Gharoun, F. Momenifar, F. Chen, and A. H. Gandomi, "Meta-learning approaches for few-shot learning: A survey of recent advances," *ACM Comput. Surveys*, vol. 56, no. 12, pp. 1–41, Dec. 2024.

[23] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[24] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[25] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[26] H. Ge et al., "Cross attention-based multi-scale convolutional fusion network for hyperspectral and LiDAR joint classification," *Remote Sens.*, vol. 16, no. 21, p. 4073, Oct. 2024.

[27] J. Cai et al., "A novel graph-attention based multimodal fusion network for joint classification of hyperspectral image and LiDAR data," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123587.

[28] D. Shen, H. Hu, F. He, F. Zhang, J. Zhao, and X. Shen, "Hierarchical prototype-aligned graph neural network for cross-scene hyperspectral image classification," *Remote Sens.*, vol. 16, no. 13, p. 2464, Jul. 2024.

[29] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[30] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2022.

[31] N. Kitaev, Ł. Kaiser, and A. Levskaya, "Reformer: The efficient transformer," 2020, *arXiv:2001.04451*.

[32] B. Yang, X. Wang, Y. Xing, C. Cheng, W. Jiang, and Q. Feng, "Modality fusion vision transformer for hyperspectral and LiDAR data collaborative classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17052–17065, 2024.

[33] D. G. Bullock, "Crop rotation," *Crit. Rev. Plant Sci.*, vol. 11, no. 4, pp. 309–326, Jan. 1992.

[34] J. Oh and C. Yun, "Provable benefit of cutout and CutMix for feature learning," 2024, *arXiv:2410.23672*.

[35] S. Xia, X. Zhang, H. Meng, J. Fan, and L. Jiao, "Two-stream networks for contrastive learning in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1903–1920, 2024.

[36] J. Nalepa, M. Myller, and M. Kawulok, "Hyperspectral data augmentation," 2019, *arXiv:1903.05580*.

[37] K. Han et al., "Deep semi-supervised learning for medical image segmentation: A review," *Expert Syst. Appl.*, vol. 245, Jul. 2024, Art. no. 123052.

[38] R. Jiao et al., "Learning with limited annotations: A survey on deep semi-supervised learning for medical image segmentation," *Comput. Biol. Med.*, vol. 169, Feb. 2024, Art. no. 107840.

[39] J. Li, Y. Ma, R. Song, B. Xi, D. Hong, and Q. Du, "A triplet semisupervised deep network for fusion classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540513.

[40] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[41] C. Pu, Y. Liu, S. Lin, X. Shi, Z. Li, and H. Huang, "Multimodal deep learning for semisupervised classification of hyperspectral and LiDAR data," *IEEE Trans. Big Data*, vol. 11, no. 2, pp. 821–834, Apr. 2025.

[42] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.

[43] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[44] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[45] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 1597–1607.

[46] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714, doi: 10.1109/TGRS.2023.3263511.

[47] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[48] D. Xiu, Z. Pan, Y. Wu, and Y. Hu, "MAGE: Multisource attention network with discriminative graph and informative entities for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539714, doi: 10.1109/TGRS.2022.3210398.

[49] S. Xia, X. Zhang, H. Meng, and L. Jiao, "Ternary modality contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522017, doi: 10.1109/TGRS.2024.3417011.

[50] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718, doi: 10.1109/TGRS.2023.3264235.

[51] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022.

[52] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6504205.

[53] G. Zhao et al., "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 7355–7370, 2023, doi: 10.1109/TGRS.2020.2982064.

[54] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514, doi: 10.1109/TGRS.2023.3311535.

[55] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.

[56] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532812, doi: 10.1109/TGRS.2022.3188529.

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